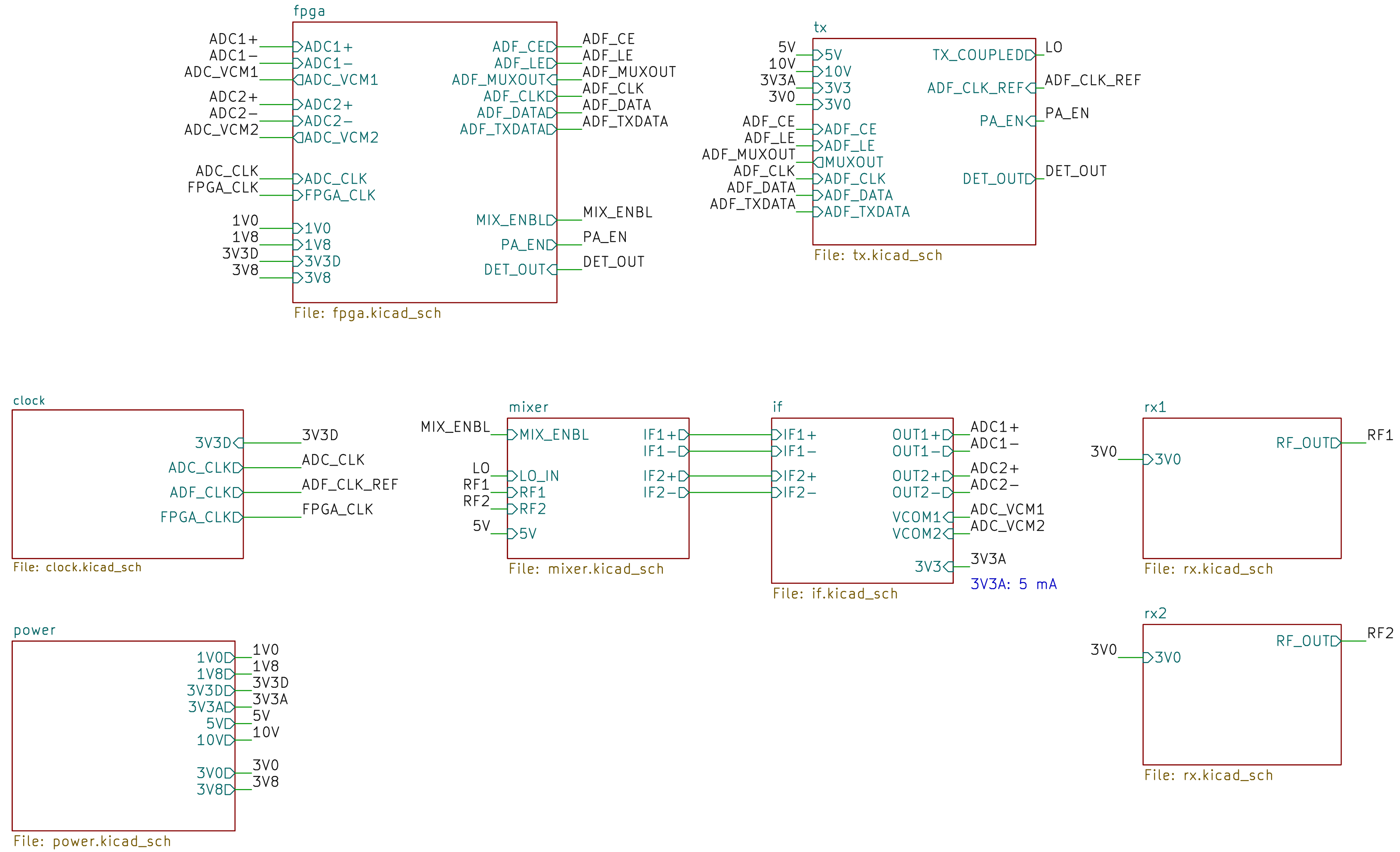
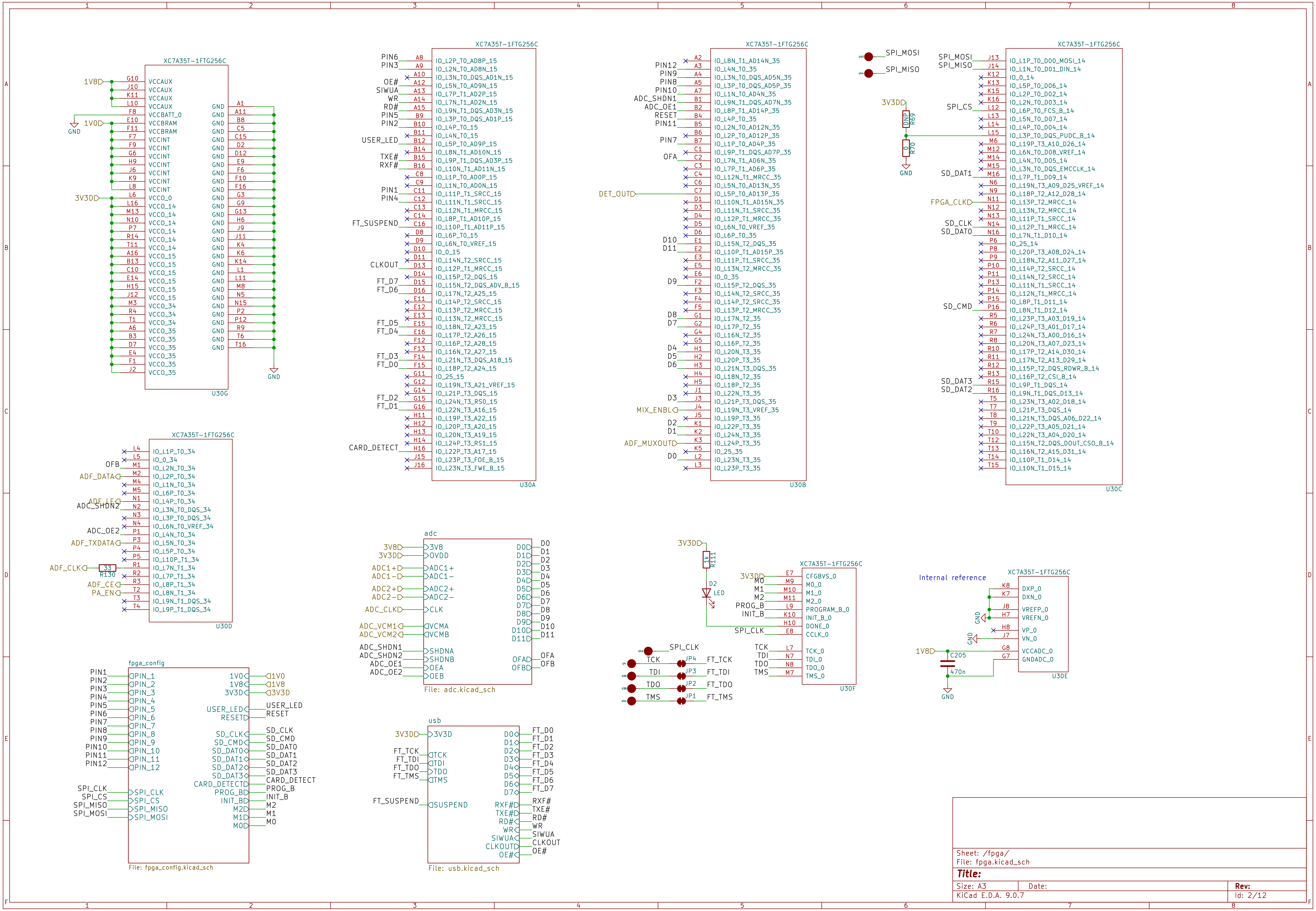


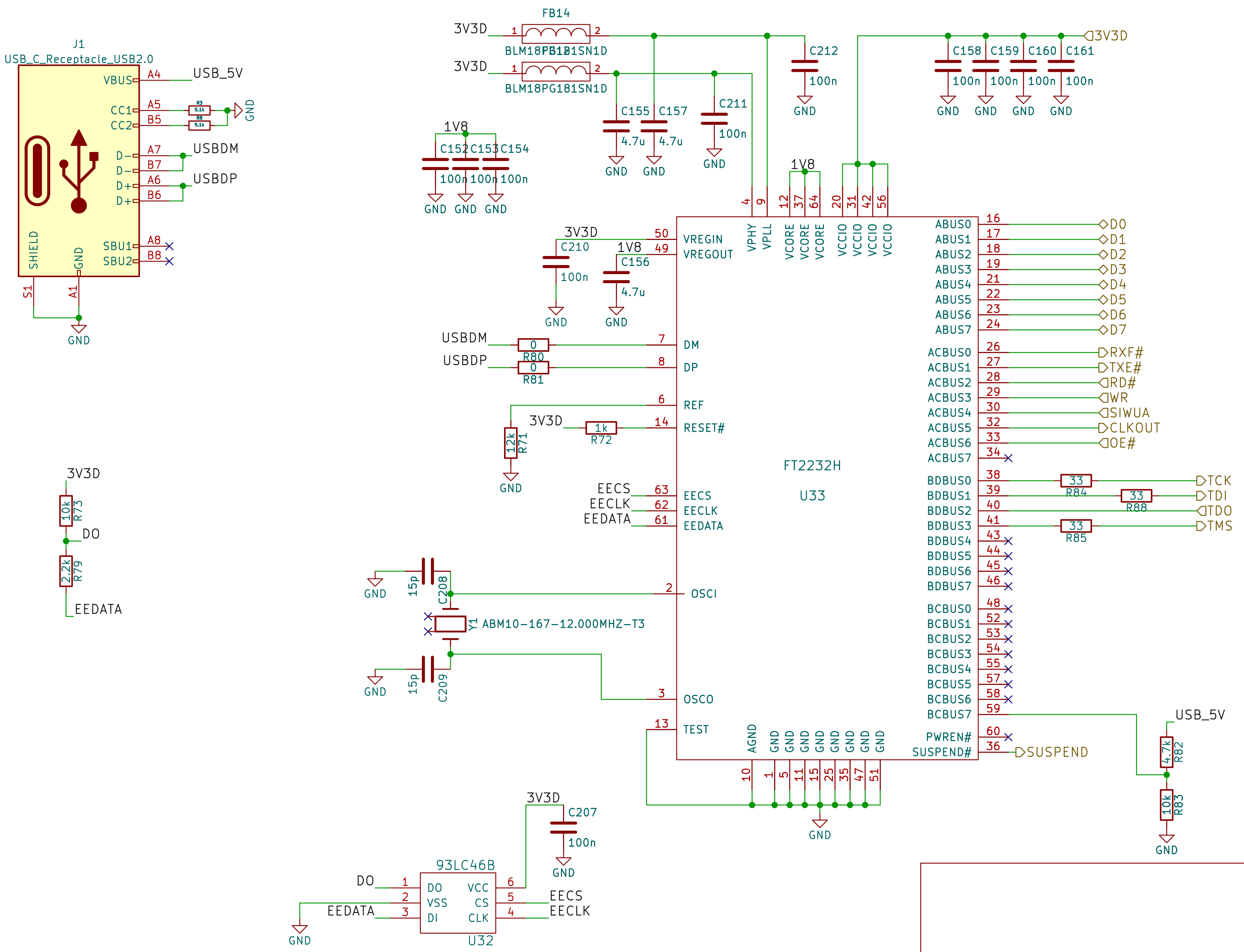
Sheet: /
File: fmcw3.kicad_sch

Title: FMCW3 by CENK KESKİN

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 1/12



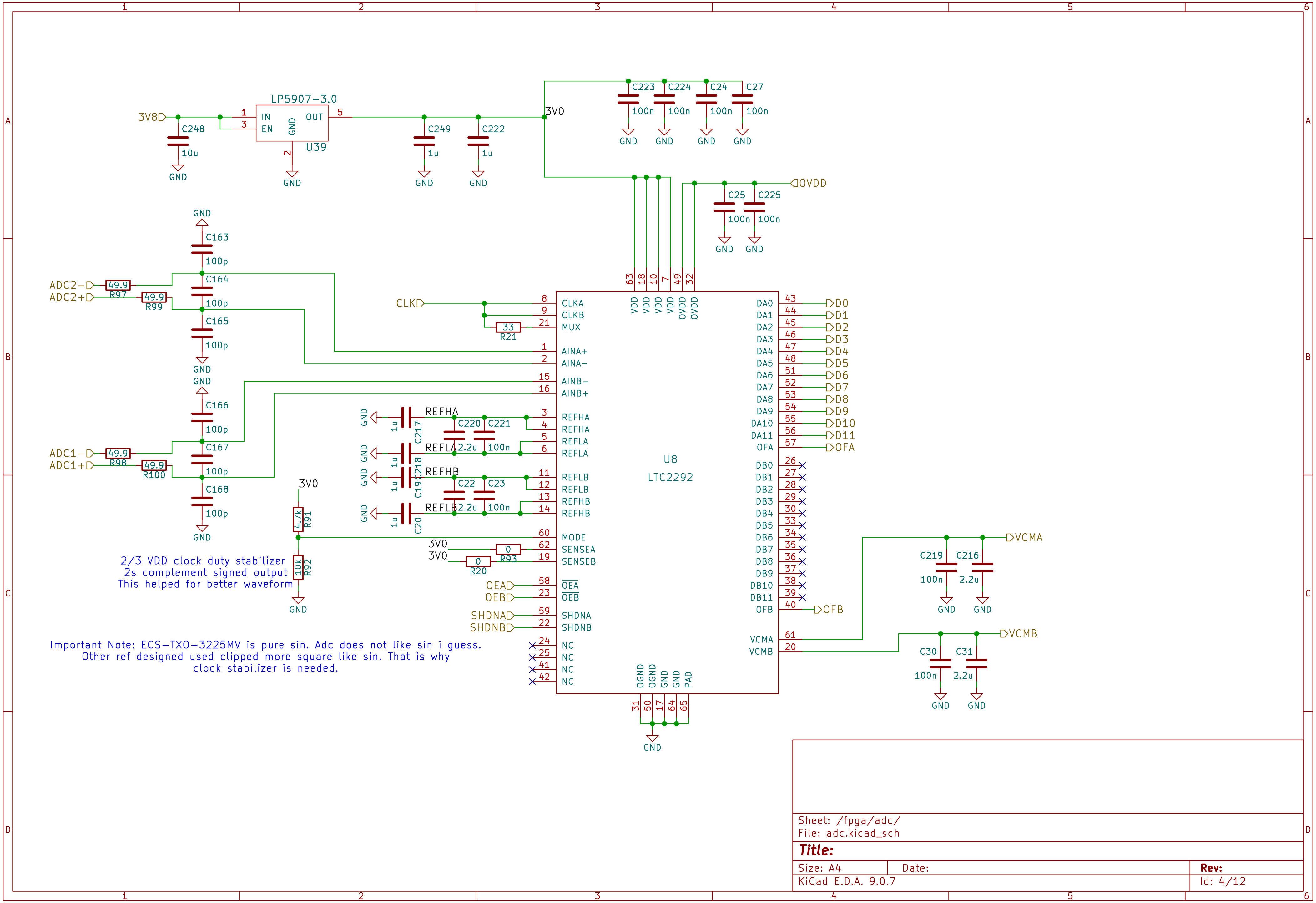
XC7A35T-1FTG256C
U30G
U30A
U30B
U30C
U30D
U30E
U30F
1V8
1V0
3V3D
GND
SPI_MOSI
SPI_MISO
SPI_CS
SPI_CLK
DNP R69
0 R70
FPGA_CLK
SD_CLK
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_CMD
DET_OUT
MIX_ENBL
ADF_MUXOUT
ADF_DATA
ADF_LE
ADF_CLK
ADF_CE
ADF_TXDATA
PA_EN
ADC_SHDN1
ADC_SHDN2
ADC_OE1
ADC_OE2
33 R130
adc
File: adc.kicad_sch
usb
File: usb.kicad_sch
fpga_config
File: fpga_config.kicad_sch
1k R111
D2 LED
PROG_B
INIT_B
M0
M1
M2
TCK
TDI
TDO
TMS
JP1
JP2
JP3
JP4
FT_TCK
FT_TDI
FT_TDO
FT_TMS
Internal reference
C205
470n
Sheet: /fpga/
File: fpga.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 2/12

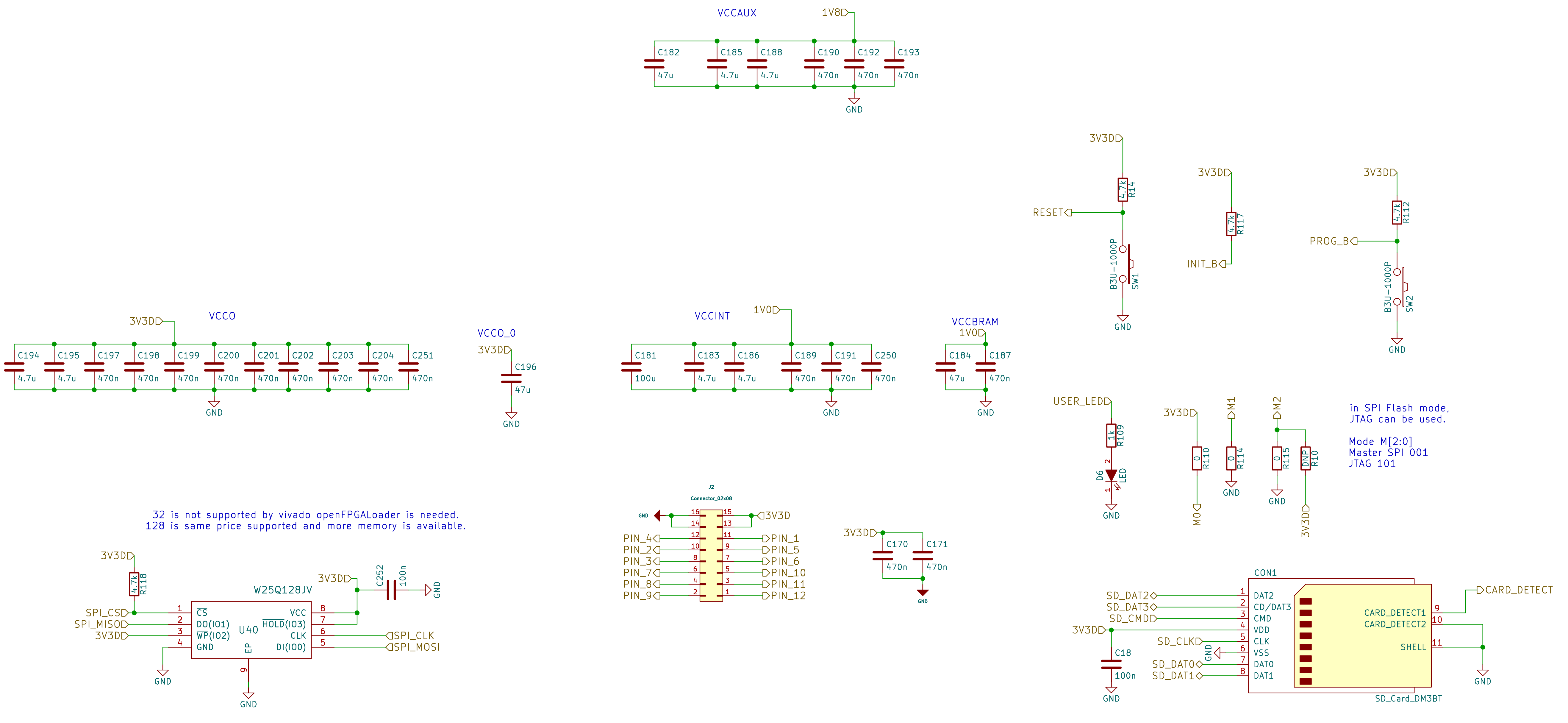


Sheet: /fpga/usb/
File: usb.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 3/12


LP5907-3.0
U39
IN
EN
OUT
GND
3V8
3V0
C248 10u
C249 1u
C222 1u
C223 100n
C224 100n
C24 100n
C27 100n
C25 100n
C225 100n
OVDD
ADC2−
ADC2+
ADC1−
ADC1+
49.9 R97
49.9 R99
49.9 R98
49.9 R100
C163 100p
C164 100p
C165 100p
C166 100p
C167 100p
C168 100p
CLK
33 R21
REFHA
REFLA
REFHB
REFLB
C217 1u
C218 1u
C19 1u
C20 1u
C220 2.2u
C221 100n
C22 2.2u
C23 100n
4.7k R91
10k R92
0 R93
0 R20
OEA
OEB
SHDNA
SHDNB
U8
LTC2292
CLKA
CLKB
MUX
AINA+
AINA−
AINB−
AINB+
MODE
SENSEA
SENSEB
NC
VDD
OVDD
OGND
GND
PAD
DA0
DA1
DA2
DA3
DA4
DA5
DA6
DA7
DA8
DA9
DA10
DA11
OFA
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
DB8
DB9
DB10
DB11
OFB
VCMA
VCMB
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
C219 100n
C216 2.2u
C30 100n
C31 2.2u
2/3 VDD clock duty stabilizer
2s complement signed output
This helped for better waveform
Important Note: ECS-TXO-3225MV is pure sin. Adc does not like sin i guess.
Other ref designed used clipped more square like sin. That is why
clock stabilizer is needed.
Sheet: /fpga/adc/
File: adc.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 4/12

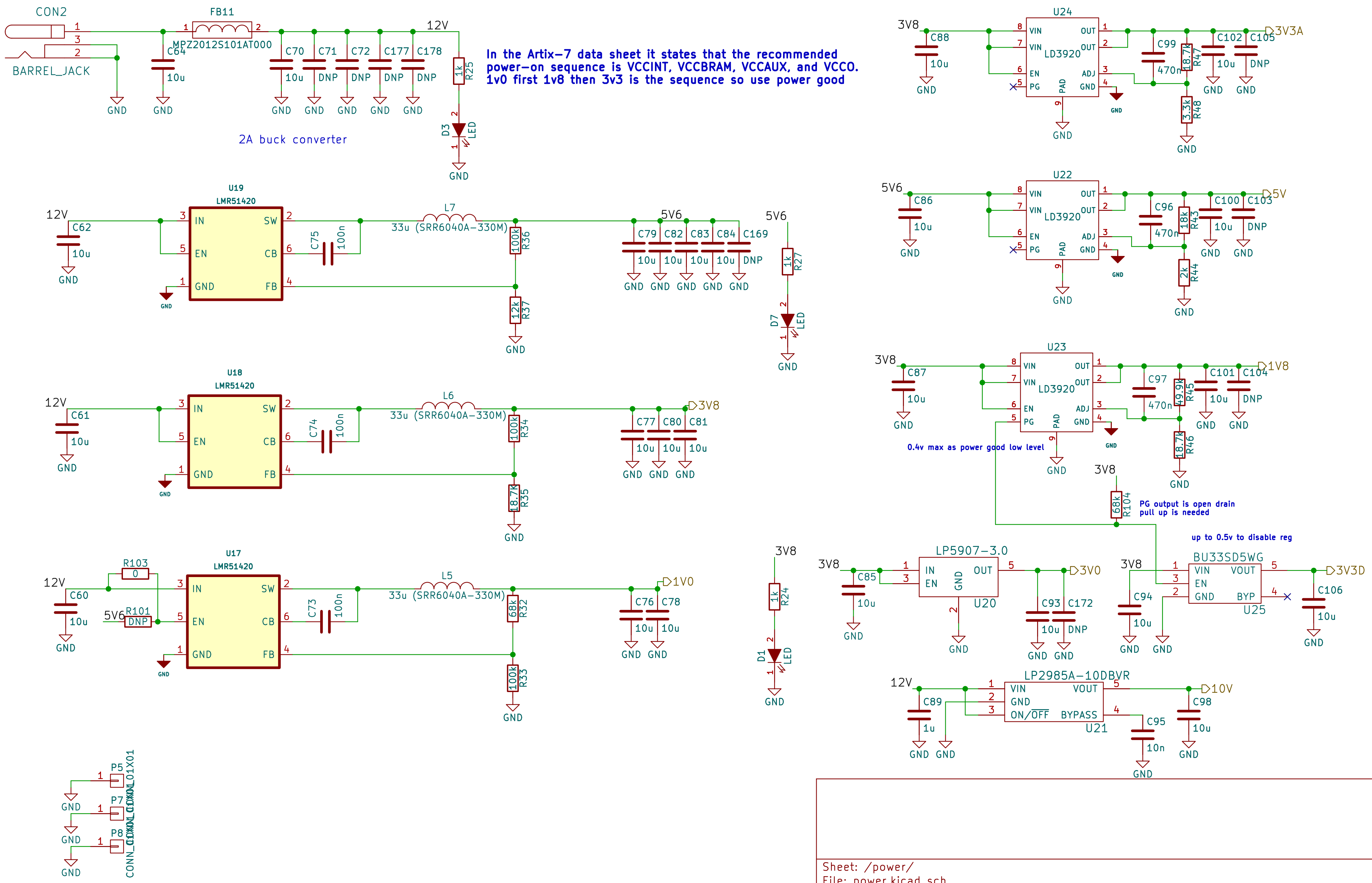


Sheet: /fpga/fpga_config/
File: fpga_config.kicad_sch

Title:

Size: A3 | Date: | Rev:

KiCad E.D.A. 9.0.7 | Id: 12/12

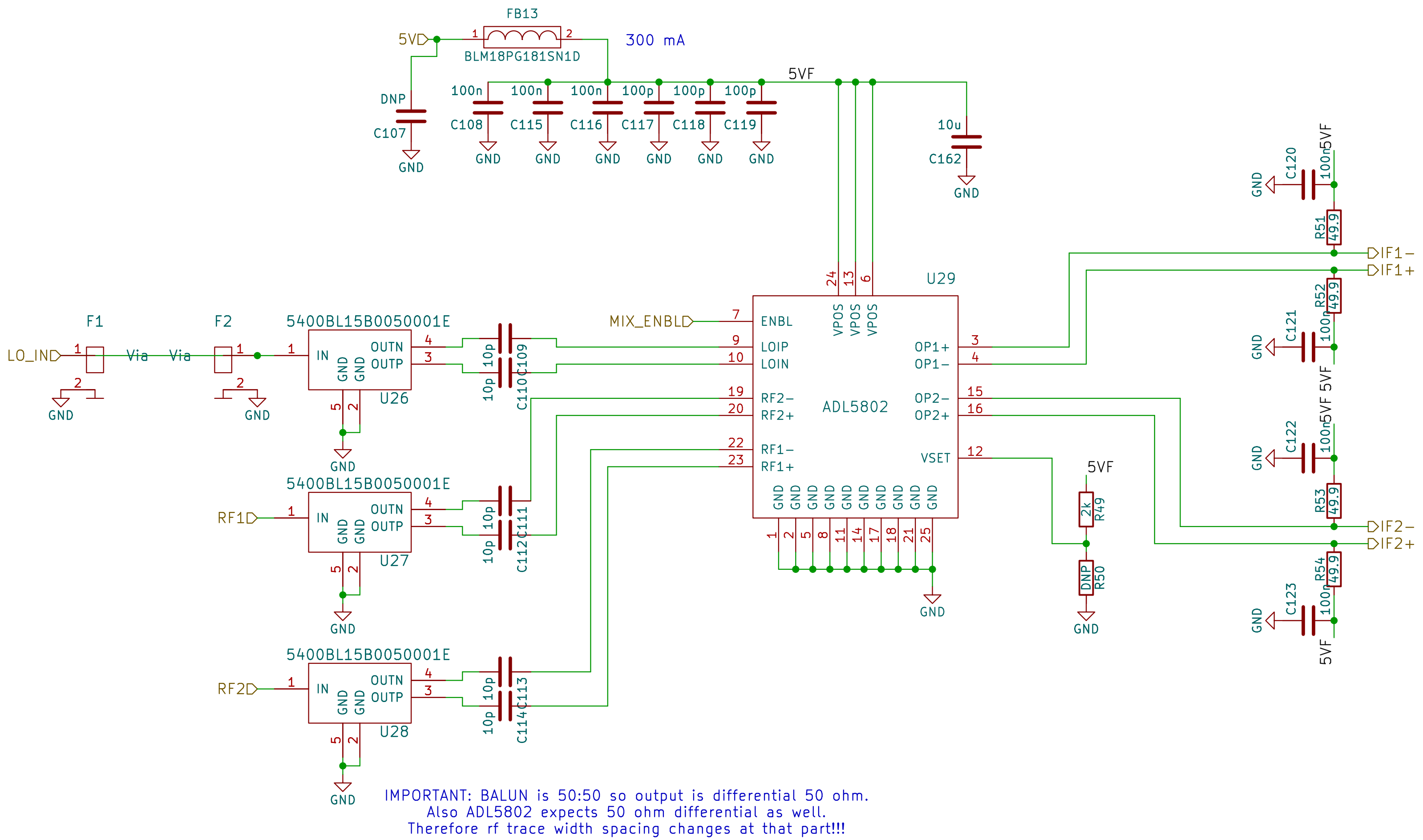


Sheet: /power/
File: power.kicad_sch

Title:

Size: A4 | Date: | Rev:

KiCad E.D.A. 9.0.7 | Id: 5/12

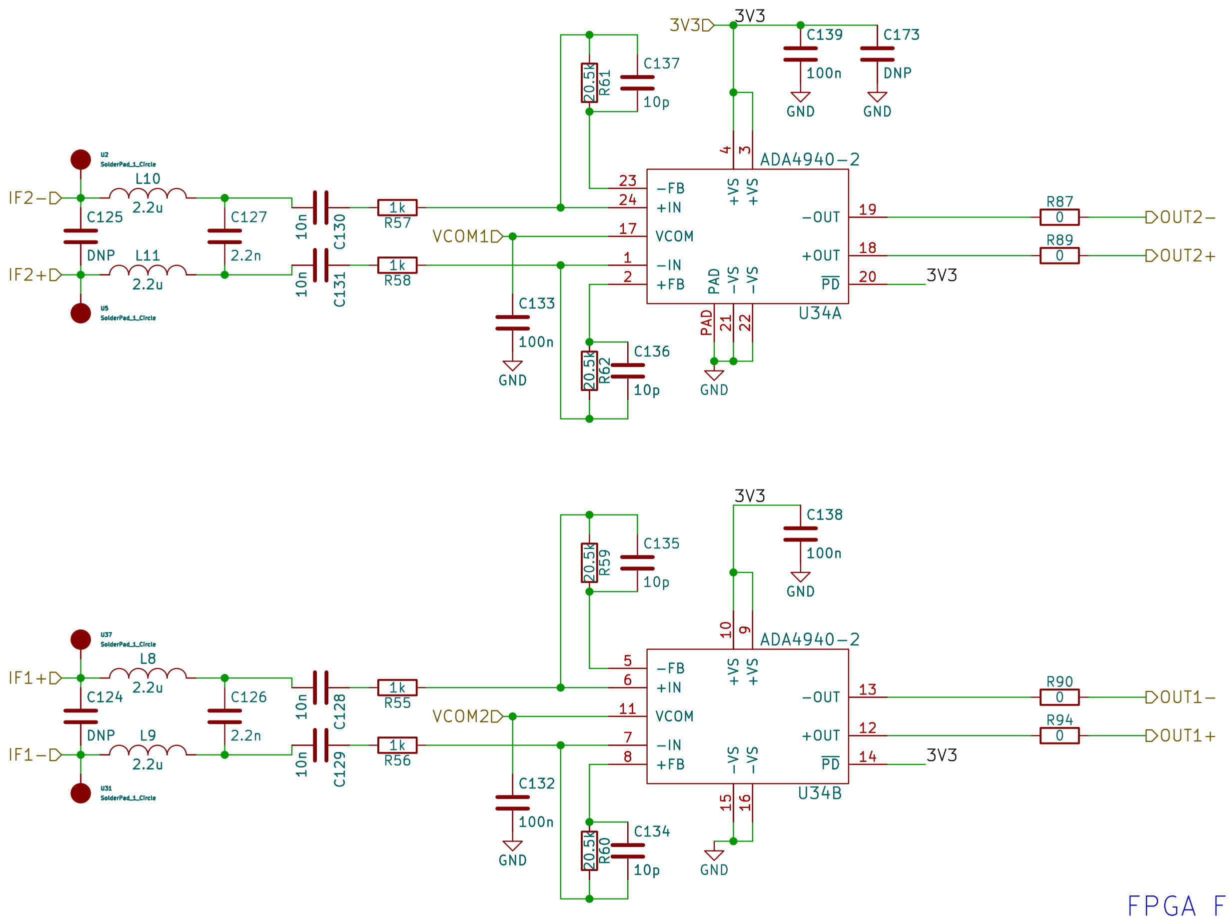


Sheet: /mixer/
File: mixer.kicad_sch

Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 9.0.7 Id: 6/12

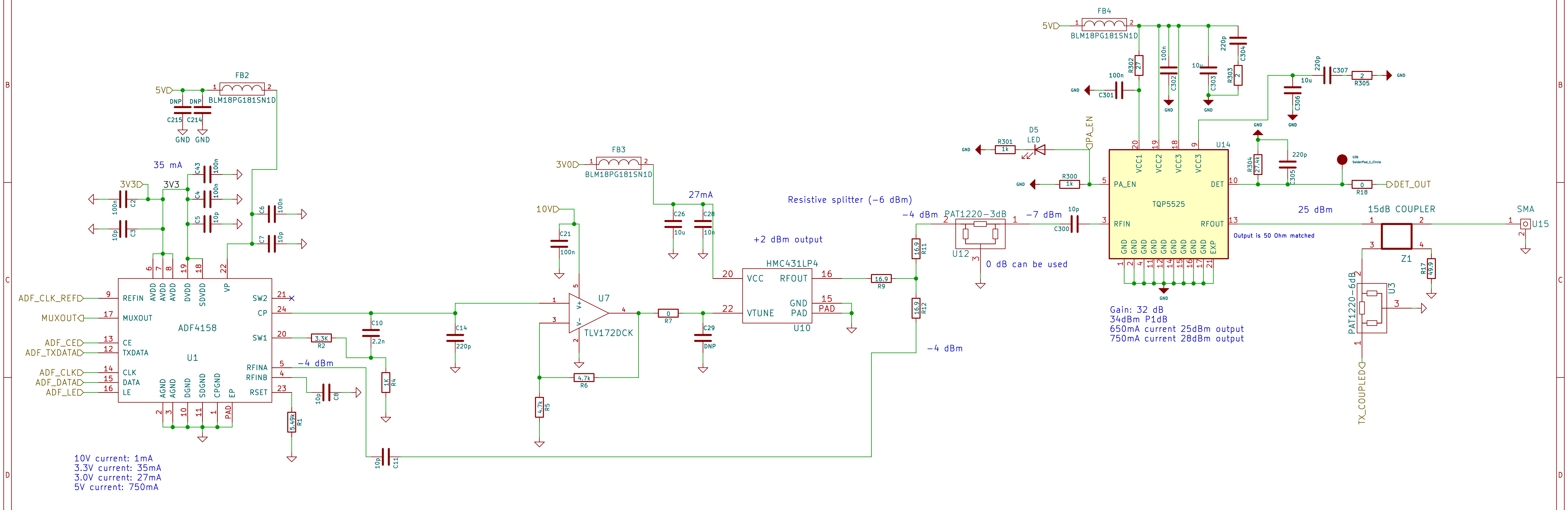


FPGA FIR Filter works perfectly.
Second stage is removed

Sheet: /if/
File: if.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 7/12

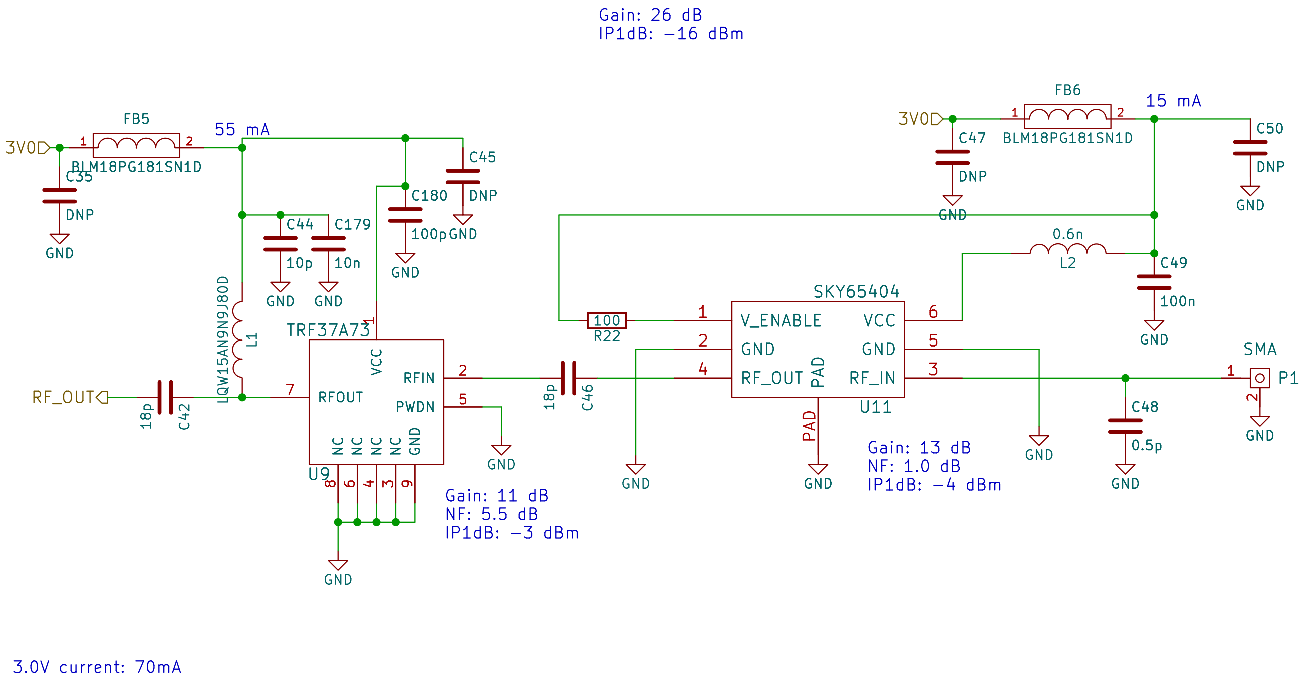


Sheet: /tx/
File: tx.kicad_sch

Title:

Size: A3 | Date: | Rev:

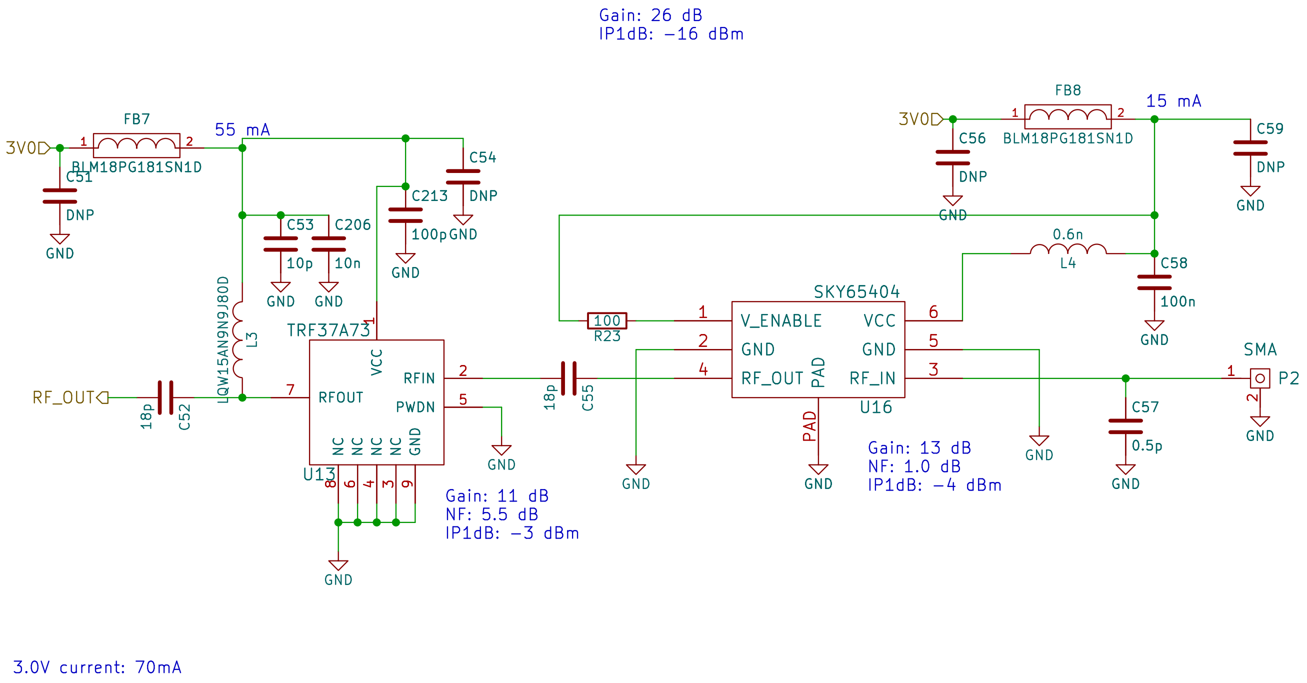
KiCad E.D.A. 9.0.7 | Id: 8/12



Sheet: /rx1/
File: rx.kicad_sch

Title:

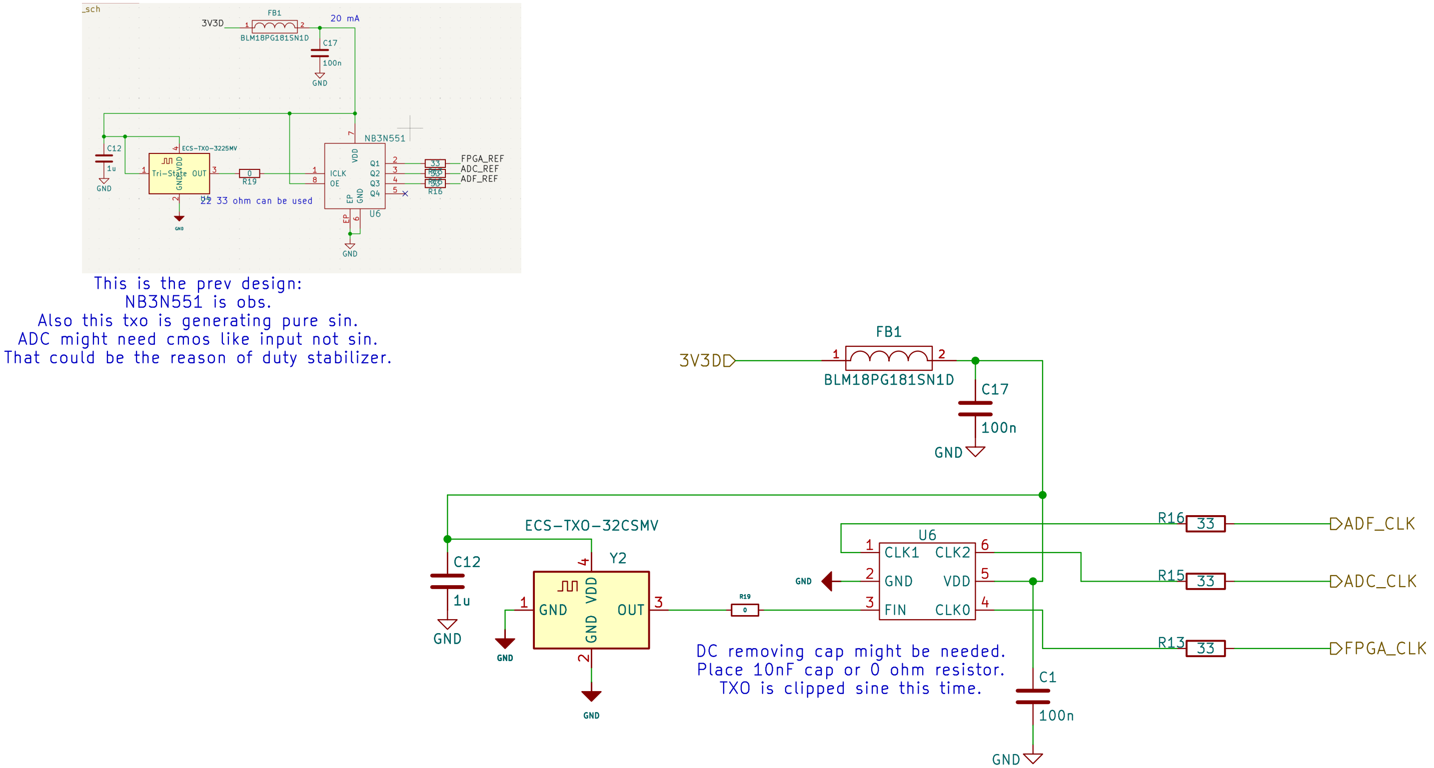
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 9/12



Sheet: /rx2/
File: rx.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 10/12



Sheet: /clock/
File: clock.kicad_sch

Title:

Size: A4 | Date: | Rev:

KiCad E.D.A. 9.0.7 | Id: 11/12